Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/13/2012
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Chung Law's email is cklaw@princeton.edu		
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2009	**End Date:**	09/30/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Project to be extended to 9/30/2014 per D. Stocker/GRC (Ed., 7/20/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Tse, Stephen (Rutgers University)		
Grant/Contract No.:	NNX09AR03A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the GRC and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the		

	identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
Task Progress:	The Requirements Definition Review (RDR) was passed in May 2010. Continual updates to the ACME Science Requirements Documents (SRD), based in part on recommendations from the Science Panel, have been and are being made based on on-going research. Specifically, the test matrix is being refined, in terms of specific parameter values and number of cases. All cases are being computationally simulated to evaluate the time and conditions at extinction, both radiative and kinetic. Summaries of the on-going ground-based investigation follow. The formulation of Taylor on the self-similar propagation of an expanding spherical piston with constant velocity was extended to an instability-wrinkled deflagration front undergoing acceleration with RF is proportional to talpha, where RF is the instantaneous flame radius, t the time, and alpha a constant exponent. The formulation describes radial compression waves pushed by the front, trajectories of gas particles, and the explosion condition in the gas upstream of the front. The instant and position of explosion are determined for a given reaction mechanism. For a step-function induction time, analytic formulas for the explosion time and position are derived, showing their dependence on the reaction and flow parameters including thermal expansion, specific heat ratio, and acceleration of the front. The state of research on developing fuel options for next-generation chemical propulsion is reviewed for aviation fuels and energetic fuels. For aviation fuels, the development is based on considerations of cost, energy security, and climate change, with Fischer–Tropsch synthetic fuels and biofuels hold potential as alternative aviation fuels. The need for basic research to develop predictive capability for the oxidative chemistry of evolving fuels in evolving engine designs is emphasized, illustrated by the intricate reaction pathways and the enormity of the reaction mechanisms involved. Recent research activities toward achieving the goal of fuel design are discussed through the development of detailed mechanisms, reduced mechanisms, and surrogate fuels. For the development of high energy-density propellants, advances in several classes of materials are discussed, including metallized and hypergolic propellants and propellants with strained and functionalized molecules, as well as nanoparticle addition. The impact of the recent progress in chemical synthesis, materials science, and nano science on these advances is noted. Microgravity experiments were conducted in the 2.2-s drop tower and zero-gravity facility at NASA-GRC to gain fundamental understanding of the effects of spinning on an otherwise spherical diffusion flame. The flames were generated by injecting either a fuel or an oxidizer mixture from a porous burner to a controlled ambient of either an oxidizer or fuel mixture, respectively. Results show that the polar flame location scales with the angular velocity monotonically as xa, where a is greater and smaller than unity for small and large spinning velocities, respectively. On the contrary, the equatorial flame location responds nonmonotonically to increasing spinning velocity: first increasing and then decreasing. The experimental observations agree well with the computational simulation where the simulated results demonstrate that the nonmonotonic response of the equatorial flame location is caused by dilution of the reactant concentration in the outwardly-directed radial flow by the product and inert that are carried by the inwardly-directed polar flow upon traversing the flame segment in the polar region. A complementary low-pressure combustion chamber is used to examine the flame structure of low-buoyancy, inverted, spherical diffusion flames, using laser-based diagnostics. The effect of buoyancy is minimized in a low-pressure environment, typically down to 0.1 atmosphere, beyond which kinetic effects would become severe. However, since the flames examined have centimeter-order diameters, buoyancy can still be significant due to the large dimension. Therefore, to further reduce buoyancy, an additional arrangement is implemented, which involves injecting the oxidizer gas into an environment of low molecular weight fuel such as hydrogen and methane. Buoyancy is reduced because the hot, low-density flame sphere is now situated in an environment of low molecular weight gas. As such, spherical and rotating non-buoyant diffusion flames can be investigated, for extended durations. The chamber has water-cooled walls (to maintain constant wall temperature), 4 optical access ports (for laser diagnostics), various probe access ports (for thermocouples, radiometer sensors, oxygen sensors, etc), and modular base-plate construction for easy changing of the burner setup configuration. The cylindrical geometry and size are similar to that of the Combustion Integrated Rack, to be used in the space-based experiments. The chamber is mounted to a 3-axis positioner to enable spatial probing of the flame structure by laser diagnostics. In terms of in-situ diagnostics, the associated equipment include an injection-seeded Spectra-Physics LAB-170-10 Nd:YAG laser and Spectra Physics CSTR-G-532 Nd:YAG-pumped dye laser, an Acton SpectraPro 500i Imagning Monochromator/Spectrograph, Roper Scientific PIMAX 1300HQ-25-FO digital UV-to-near-IR ICCD camera, Hamamatsu R928 photomultiplier, Agilent 54845A Infiniium 1.5GHz digital oscilloscope, Kaiser Optical 2.0” holographic SuperNotch-Plus Filter at 532nm, and Berkeley Nucleonics Model 555-4C 4 channel digital delay generator. The system is used for spontaneous Raman spectroscopy (SRS), laser-induced fluorescence (LIF), and filtered Rayleigh scattering (FRS) to probe the flame for gas-phase temperature and major/radical gaseous species concentrations. Particle imaging velocimetry (PIV) is utilized to map the aerodynamic flow field. As such the controlling mechanisms (for 1D and 2D axisymmetric spinning flames) can then be unambiguously identified, analyzed, modeled, and compared with detailed computational simulation for quantitative realism, fundamental understanding, and predictive capability.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Akkerman V, Law CK, Bychkov V. "Self-similar accelerative propagation of expanding wrinkled flames and explosion triggering." Physical Review E, 2011 Feb;83(2):026305. http://dx.doi.org/10.1103/PhysRevE.83.026305 , Feb-2011

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Articles in Peer-reviewed Journals	Law CK. "Fuel options for next generation chemical propulsion" AIAA Journal, 2012 Jan;50(1):19-36. http://dx.doi.org/10.2514/1.J051328 , Jan-2012